

# RETAIL ASSORTMENT INTELLIGENCE: USING MACHINE LEARNING FOR SMARTER INVENTORY DECISIONS

## 01 Introduction

In the retail industry, managing inventory efficiently is one of the biggest challenges businesses face. Overstocking products can increase storage costs and waste, while understocking can lead to missed sales opportunities and dissatisfied customers. As retail markets become more competitive and customer demand becomes harder to predict, businesses increasingly rely on data analytics and machine learning to improve inventory planning and decision-making.



Our capstone project, Retail Assortment Intelligence: ABC Analysis & Demand Forecasting for Inventory Optimization, focuses on using retail transaction data to better understand product performance, forecast future demand, and generate actionable business insights. By combining exploratory data analysis, ABC product segmentation, machine learning forecasting models, and interactive dashboarding, we aimed to build a practical business intelligence solution that supports smarter inventory management.

## 02 The Problem

Traditional inventory management methods often rely on manual estimation, historical intuition, or simple reporting techniques. While these methods may work for smaller businesses, they become increasingly inefficient when handling thousands of products and large volumes of transactional data.

Retail businesses commonly face challenges such as:

- Overstocking low-performing products
- Running out of high-demand items
- Poor visibility into seasonal demand trends
- Difficulty identifying which products contribute most to revenue
- Inefficient stock allocation across product categories

To address these challenges, businesses need a more data-driven approach that combines historical sales analysis with predictive modeling.

## 03 Our Approach

To solve this problem, we developed a retail analytics pipeline that combines:

- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- ABC Product Segmentation
- Machine Learning Forecasting
- Interactive Power BI Dashboards
- Business Recommendation Generation

The goal was not only to analyze historical retail data, but also to transform the analysis into practical recommendations that businesses can directly use to improve inventory planning and operational efficiency.

## 04 Dataset Overview

For this project, we selected the **Online Retail Dataset**, which contains transactional records from a UK-based online retailer between December 2010 and December 2011.

The dataset initially contained over **541,909 retail transactions** and included important fields such as:

- Invoice Date
- Product Description
- Quantity Sold
- Unit Price
- Revenue
- Country
- Customer ID

```
str(df)

tibble [541,909 × 8] (S3: tbl_df/tbl/data.frame)
 $ InvoiceNo : chr [1:541909] "536365" "536365" "536365" ...
 $ StockCode : chr [1:541909] "8512284" "71853" "844868" "844290" ...
 $ Description: chr [1:541909] "WHITE HANGING HEART T-LIGHT HOLDER" "WHITE METAL LANTERN" "CREAM CUPID HEARTS COAT HANGER" "KNITTED UNION FLAG HOT WATER BOTTLE" ...
 $ Quantity : num [1:541909] 6 6 8 6 6 2 6 6 6 32 ...
 $ InvoiceDate: POSIXct[1:541909], format: "2010-12-01 08:26:00" "2010-12-01 08:26:00" ...
 $ UnitPrice : num [1:541909] 2.55 3.39 2.75 3.39 3.39 7.65 4.25 1.85 1.85 1.69 ...
 $ CustomerID: num [1:541909] 17850 17850 17850 17850 17850 ...
 $ Country : chr [1:541909] "United Kingdom" "United Kingdom" "United Kingdom" "United Kingdom" ...
```

We initially explored two datasets before deciding to continue with the Online Retail dataset because it contained richer transactional information suitable for forecasting and assortment planning.

The dataset allowed us to:

- Analyze customer purchasing behavior
- Identify important products
- Detect seasonal demand patterns
- Forecast future product demand

```
summary(df)

InvoiceNo      StockCode      Description      Quantity
Length:541909 Length:541909 Length:541909 Min. :~80995.000
Class :character Class :character Class :character 1st Qu.: 1.000
Mode :character Mode :character Mode :character Median : 3.000
Mean : 9.552
3rd Qu.: 10.000
Max. : 80995.000

InvoiceDate      UnitPrice      CustomerID
Min. :2010-12-01 08:26:00 Min. :~11862.060 Min. :12346
1st Qu.:2011-03-28 11:34:00 1st Qu.: 1.250 1st Qu.:13953
Median :2011-07-19 17:17:00 Median : 2.080 Median :15152
Mean :2011-07-04 13:34:57 Mean : 4.611 Mean :15288
3rd Qu.:2011-10-19 11:27:00 3rd Qu.: 4.130 3rd Qu.:16791
Max. :2011-12-09 12:50:00 Max. : 38970.000 Max. :18287
NA's :135080

Country
Length:541909
Class :character
Mode :character
```

05

Data Cleaning & Preparation

Before beginning the analysis, the dataset required significant cleaning and preprocessing.

The cleaning process included:

- Removing missing Customer IDs
- Removing cancelled transactions
- Filtering negative or zero quantity values
- Filtering invalid pricing records
- Standardizing date and time variables

After cleaning, the dataset size was reduced from approximately **541K rows to 392K valid transactions**.

We then performed feature engineering by creating:

- Revenue variables
- Year and Month features
- Weekday labels
- Time-of-day categories
- Lag features for forecasting
- Rolling averages for demand smoothing

These engineered features later became important inputs for the machine learning forecasting models.

06

Exploratory Data Analysis (EDA)

Exploratory Data Analysis helped us better understand the structure and behavior of the retail data before applying machine learning techniques.

During the analysis, we explored:

- Revenue trends over time
- Product sales distributions
- Seasonal demand patterns
- Customer purchasing behavior
- Operational demand patterns

One of the most important findings from the EDA stage was the strong seasonal increase in demand during **November and December**, which aligns with the holiday shopping season.

We also observed clear purchasing patterns throughout the week, with stronger demand occurring during weekday afternoons.

These findings demonstrated that retail demand is not random, but instead follows identifiable operational and seasonal trends that businesses can use for inventory planning.

07

ABC Product Segmentation

One of the core analytical techniques used in this project was **ABC Product Segmentation**.

ABC analysis is based on the **Pareto Principle**, also known as the **80/20 rule**, which states that a relatively small percentage of products often generates the majority of the revenue.

Using cumulative revenue contribution, products were divided into three categories:

| Category | Description           | Revenue Contribution |
|----------|-----------------------|----------------------|
| A        | High-value products   | ~80%                 |
| B        | Medium-value products | ~15%                 |
| C        | Low-value products    | ~5%                  |

The results clearly showed that a small group of Category A products generated most of the total revenue, while Category C products represented a large number of low-performing items.

This segmentation provided important business value by helping prioritize:

- High-performing products
- Inventory investment decisions
- Stock allocation strategies
- Product assortment planning
- ABC analysis also became an important feature in our forecasting models and business recommendation system.



## 08 Machine Learning Forecasting

To forecast future product demand, we developed multiple machine learning models using historical sales data. The models included:

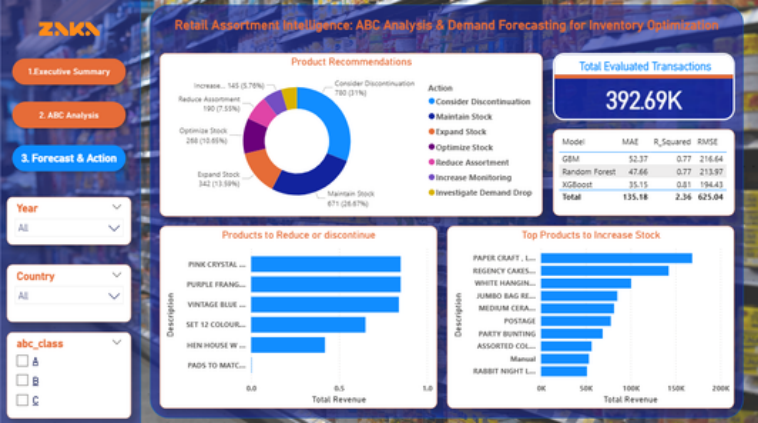
- Linear Regression
- Random Forest
- Gradient Boosting
- XGBoost

The forecasting models used features such as:

- Historical quantities sold
- Average product price
- ABC category
- Lag variables
- Rolling averages
- Order counts

We also applied:

- Log transformations to reduce skewness
- Time-based train/test splitting
- Model evaluation metrics such as RMSE and R<sup>2</sup>



Among all models tested, XGBoost achieved the strongest performance, reaching an R<sup>2</sup> score of approximately 0.81, making it the most accurate forecasting model in our project. The forecasting system allowed us to estimate future product demand and support smarter inventory planning decisions.

## 09 Business Insights & Findings

The analytical and forecasting stages generated several valuable business insights.

### 1. Inventory Optimization

The forecasting models helped identify products with:

- High predicted demand
- Low predicted demand

Products with strong forecasted demand were recommended for:

- Stock expansion
- Higher inventory prioritization
- Increased monitoring during peak seasons

Meanwhile, products with consistently low predicted demand were recommended for:

- Reduced assortment
- Inventory optimization
- Possible discontinuation

This helps businesses reduce inventory waste while protecting high-performing products from stock shortages.

### 2. Seasonal Demand Trends

The analysis revealed strong seasonal demand peaks around **November and December**.

This suggests that businesses should:

- Increase inventory planning approximately 4–6 weeks before peak periods
- Prepare for higher operational activity during Q4
- Improve supply chain coordination before demand surges

Understanding seasonal demand trends is critical for reducing stock shortages during high-revenue periods.

### 3. Operational Demand Patterns

We also analyzed customer purchasing behavior by:

- Day of week
- Time of day

The results showed that:

- Demand was generally stronger during weekday afternoons
- Some weekdays consistently outperformed others in total sales volume

These insights can support:

- Staffing decisions
- Warehouse planning
- Operational scheduling
- Inventory replenishment timing

## 10 Power BI Dashboard

To make the results more interactive and business-friendly, we developed a Power BI dashboard that visualizes the project findings.

The dashboard includes:

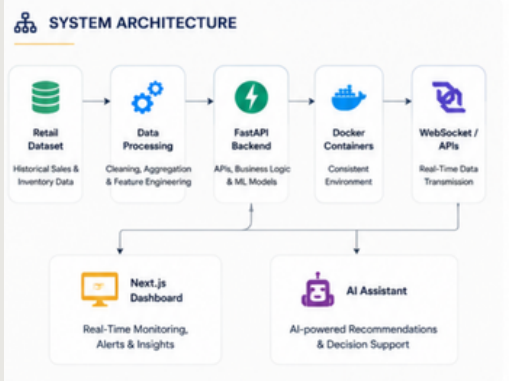
- Executive summary KPIs
- ABC revenue contribution analysis
- Product segmentation visuals
- Forecasting insights
- Product action recommendations
- Interactive filters by year, country, and ABC category

The dashboard transforms technical analysis into an accessible business intelligence tool that decision-makers can use to monitor product performance and inventory strategies in real time.

## 11 Prototype Development

To transform our analytical findings into a practical business solution, we developed a real-time inventory optimization prototype that simulates a modern retail decision-support system. The prototype was designed and developed using Visual Studio Code as the primary development environment, with Docker utilized to containerize the application and ensure a consistent and scalable deployment process.

The system architecture consists of a FastAPI backend responsible for data processing, business logic, and real-time data simulation, and a Next.js frontend that delivers an interactive user experience. Communication between the frontend and backend is enabled through APIs and WebSocket connections, allowing the platform to display continuously updated inventory records and alerts.



Since access to a live retail environment was not available, a simulated real-time data stream was implemented, generating new inventory records every few seconds to mimic operational retail activities. This allowed the prototype to demonstrate how inventory risks, stock movements, and product performance can be monitored in real time. The prototype integrates several key capabilities, including inventory monitoring, demand forecasting insights, automated alert generation, inventory optimization recommendations, and an AI-powered assistant that helps users interpret data and support decision-making. By combining advanced analytics with an interactive user interface, the prototype showcases how data-driven inventory management can improve operational efficiency and support smarter retail decisions.

### Key Features of the Prototype

#### 1.Real-Time Monitoring

The system continuously simulates inventory transactions, providing live visibility into stock movements, inventory levels, and product performance.

#### 2.Automated Alert Management

Intelligent alerts are generated to identify critical inventory situations such as low stock levels, excess inventory, and potential demand fluctuations.

#### 3.Demand Forecasting Insights

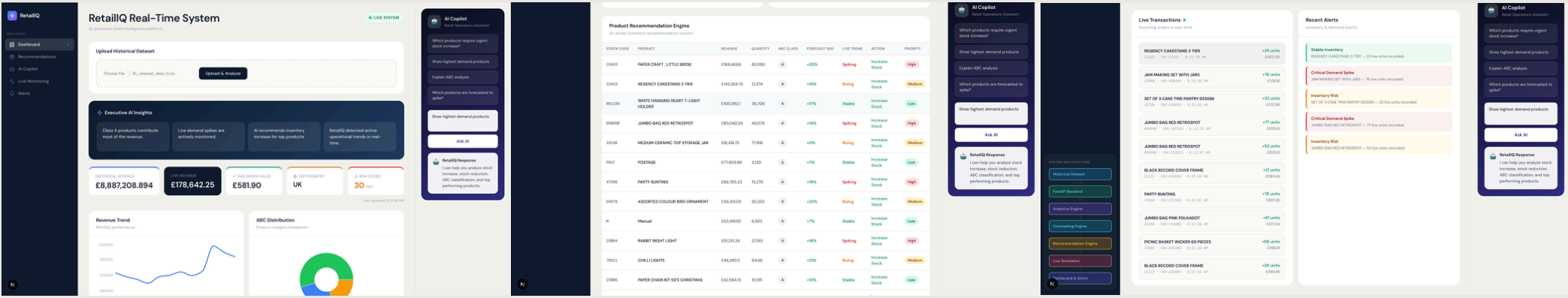
Machine learning forecasting models provide future demand trends, helping businesses make proactive inventory planning decisions.

#### 4.Inventory Optimization Recommendations

The prototype analyzes inventory performance and generates data-driven recommendations to improve stock allocation and reduce operational inefficiencies.

#### 5.AI-Powered Decision Support

An integrated AI assistant enables users to interact with inventory data through natural language queries and receive actionable business insights.



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## *Business Impact*

This project demonstrates how data analytics and machine learning can directly support retail decision-making.

The final solution can help businesses:

- Reduce inventory waste
- Improve stock planning
- Prevent stock shortages
- Prioritize high-performing products
- Improve operational efficiency
- Support data-driven assortment strategies

By combining ABC analysis with demand forecasting, businesses can move away from intuition-based inventory decisions and adopt smarter, evidence-based planning strategies.

## *Limitations*

Although the project achieved strong results, several limitations still exist.

Some limitations include:

- Dependence on historical transaction data only
- Limited external variables such as promotions or economic conditions
- Forecasting uncertainty during unexpected market changes
- Limited real-time integration capabilities

These limitations create opportunities for future improvement and expansion.

## *Future Work*

Several future improvements can further strengthen the project, including:

- Real-time forecasting integration
- Supplier and warehouse optimization
- Automated inventory alerts
- Deep learning forecasting models
- Customer segmentation integration
- Live dashboard deployment

Future versions of the project could also include real-time retail streams and cloud-based deployment for enterprise-level scalability

## *Final Thoughts*

Retail inventory optimization is no longer just an operational task — it is becoming a strategic business function powered by data analytics and machine learning.

Through this project, we demonstrated how combining:

- ABC segmentation,
- machine learning forecasting,
- and interactive dashboards

can generate meaningful business insights and support smarter inventory decisions.

By transforming raw retail transaction data into actionable intelligence, businesses can improve operational efficiency, reduce waste, and maximize revenue opportunities in increasingly competitive retail markets.

## *Tools & Technologies Used*

- Google Colab
- R Programming Language
- Power BI
- XGBoost
- Random Forest
- tidyverse
- ggplot2
- Visual Studio
- Docker

## *References*

Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.

Pareto, V. (1906). Manual of Political Economy.  
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